00 — Squadron Colby: Team and Roles

Mission: 03 — Multi-Head Attention, RoPE Vector Rotation & Grouped-Query KV-Cache Optimization
Course: Conversational AI: History, Applications, Future · WiSe 2026/27 · BTU Cottbus-Senftenberg
Repository: https://github.com/xxrafidxx/Squadron-Colby---Conversational-AI

Team

Name	Squadron station	Mission 03 role	Deliverable
Rafid Mohammad Ahsan	Core & Lead Architect	Part 1 · Systems Architect (Lead Engineer)	`01_MultiHead_Attention_Core.py`, `ps03_solution.ipynb`
Rafid Mohammad Ahsan	Protocol & Interface Specialist	Part 2 · Communications & Protocol Officer	`02_RoPE_Positional_Engine.py`
Tahjib Al Sakin	Data & Memory Engineer	Part 3 · Data & Payload Specialist	`03_GQA_and_KVCache_Benchmark.py`
Kunal Choudhary	Security & Ethical Navigator	Part 4 · Security & Ethics Sentinel	`04_Memory_Hierarchy_and_Safety_Audit.md`
Mubina Rahaman Jerin	AI Security Officer	EU AI Act compliance	`06_AI_Act_Compliance.md`

Who did what

Name	Work done
Rafid Mohammad Ahsan	*Architect:* built `MultiHeadSelfAttention` in pure PyTorch (W_Q, W_K, W_V, W_O, head split/merge, hand-written softmax, causal and padding masks) with a variance-preservation test and a softmax-saturation study; assembled the notebook and repository. *Protocol:* implemented RoPE as complex 2D rotation, verified norm preservation and relative invariance, compared RoPE with ALiBi (curves, heatmaps) and analysed context extrapolation with Position Interpolation.
Tahjib Al Sakin	Built `GQAAttention` (grouped queries without copying K/V) and a stateful `KVCache`; proved cached decoding equals the full pass; benchmarked MHA, GQA 4:1 and MQA 32:1 at 1k–32k tokens for cache size, bandwidth at 50 tok/s and per-token latency.
Kunal Choudhary	Derived standard vs FlashAttention HBM traffic; simulated a KV-cache denial-of-service attack on an H100 80 GB serving LLaMA-3-8B, finding the CUDA-OOM point; evaluated PagedAttention and DeepSeek's MLA as defences.
Mubina Rahaman Jerin	Reviewed the mission against the EU AI Act (AI-system definition, GPAI duties, Art. 15 and Art. 55 robustness and cybersecurity) and listed deployment safeguards.

Moodle submission (3 files): `00_Team_Roles.pdf` · `Mission03_Role_Deliverables_01-04.pdf` (Parts 1–4, one section per role) · `06_AI_Act_Compliance.pdf`. Each role document is also a separate PDF in the repository (`mission-03/pdf/`).